

Chapter1

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1 Introduction

Def 1. If A is any set (whose elements may be numbers or any other objects), we write $x \in A$ to indicate that x is a member (or an element) of A . If x is not a member of A , we write $x \notin A$.

The set which contains no element will be called the *empty set*. If a set has at least one element, it is called *nonempty*.

If A and B are sets, and if every element of A is an element of B , we say that A is a subset of B , and we write $A \subset B$, or $B \supset A$. If, in addition, there is an element of B which is not in A , then A is said to be a *proper* subset of B . Note that $A \subset A$ for every set A .

If $A \subset B$ and $B \subset A$, we write $A = B$. Otherwise, $A \neq B$.

Def 2. The set of all rational numbers will be denoted by $\mathbb{Q}$.

2 Ordered Sets

Def 3. Let S be a set. An *order* on S is a relation, denoted by $<$, with the following two properties

(i) If $x \in S$ and $y \in S$, then one and only one of the statements

$$x < y, x = y, y < x$$

is true.

(ii) If $x, y, z \in S$, if $x < y$ and $y < z$, then $x < z$.

Def 4. An *ordered set* is a set S in which an order is defined.

Def 5. Suppose S is an ordered set, and $E \subset S$. If there exists a $\beta \in S$ such that $x \leq \beta$ for every $x \in E$, we say that E is *bounded above*, and call β an *upper bound* of E .

Def 6. Suppose S is an ordered set, $E \subset S$, and E is bounded above. Suppose there exists an $\alpha \in E$ with the following properties:

- (i) α is an upper bound of E .
- (ii) If $\gamma < \alpha$ then γ is not an upper bound of E .

Then α is called the *least upper bound* of E [that there is at most one such α is clear from (ii)] or the *supremum* of E , and we write

$$\alpha = \sup E.$$

The *greatest lower bound*, or *infimum*, of a set E which is bounded below is defined in the same manner: The statement

$$\alpha = \inf E$$

means that α is a lower bound of E and that no β with $\beta > \alpha$ is a lower bound of E .

Def 7. An ordered set S is said to have the *least-upper-bound property* if the following is true:

If $E \subset S$, E is not empty, and E is bounded above, then $\sup E$ exists in S .

Thm 8. Suppose S is an ordered set with the least-upper-bound property, $B \subset S$, B is not empty, and B is bounded below. Let L be the set of all lower bounds of B . Then

$$\alpha = \sup L$$

exists in S and $\alpha = \inf B$.

In particular, $\inf B$ exists in S .

3 Fields

Def 9. A *field* is a set $\mathbb{F}$ with two operations, called *addition* and *multiplication*, which satisfy the following so-called "field axioms" (A), (M), and (D):

(A) Axioms for addition

(A1) $x \in \mathbb{F}$ and $y \in \mathbb{F}$, then their sum $x + y$ is in $\mathbb{F}$.

- (A2) Addition is commutative: $x + y = y + x$ for all $x, y \in \mathbb{F}$.
- (A3) Addition is associative: $(x + y) + z = x + (y + z)$ for all $x, y, z \in \mathbb{F}$.
- (A4) $\mathbb{F}$ contains an element 0 such that $0 + x = x$ for every $x \in \mathbb{F}$.
- (A5) To every $x \in \mathbb{F}$ corresponds an element $-x \in \mathbb{F}$ such that

$$x + (-x) = 0.$$

(M) Axioms for multiplication

- (M1) If $x, y \in \mathbb{F}$, then their product xy is in $\mathbb{F}$.
- (M2) Multiplication is commutative: $xy = yx$ for all $x, y \in \mathbb{F}$.
- (M3) Multiplication is associative: $(xy)z = x(yz)$ for all $x, y, z \in \mathbb{F}$.
- (M4) $\mathbb{F}$ contains an element $1 \neq 0$ such that $1x = x$ for every $x \in \mathbb{F}$.
- (M5) If $x \in \mathbb{F}$ and $x \neq 0$ then there exists an element $1/x \in \mathbb{F}$ such that

$$x \cdot (1/x) = 1.$$

(D) The distribution law

$$x(y + z) = xy + xz$$

holds for all $x, y, z \in \mathbb{F}$.

Prop 10. The axioms for addition imply the following statements.

- (a) If $x + y = x + z$ then $y = z$.
- (b) If $x + y = x$ then $y = 0$.
- (c) If $x + y = 0$ then $y = -x$.
- (d) $-(-x) = x$.

Prop 11. The axioms for multiplication imply the following statements

- (a) If $x \neq 0$ and $xy = yz$ then $y = z$.
- (b) If $x \neq 0$ and $xy = x$ then $y = 1$.
- (c) If $x \neq 0$ and $xy = 1$ then $y = 1/x$.
- (d) If $x \neq 0$ then $1/(1/x) = x$.

Prop 12. The field axioms imply the following statements, for any $x, y, z \in \mathbb{F}$.

- (a) $0x = 0$.
- (b) If $x \neq 0$ and $y \neq 0$ then $xy \neq 0$.
- (c) $(-x)y = -(xy) = x(-y)$.
- (d) $(-x)(-y) = xy$.

Def 13. An *ordered field* is a field $\mathbb{F}$ which is also an *ordered set*, such that

- (i) $x + y < x + z$ if $x, y, z \in \mathbb{F}$ and $y < z$,
- (ii) $xy > 0$ if $x \in \mathbb{F}, y \in \mathbb{F}$, and $y > 0$.

If $x > 0$, we call x *positive*; if $x < 0$, x is *negative*.

Prop 14. The following statements are true in every ordered field.

- (a) If $x > 0$ then $-x < 0$, and vice versa.
- (b) If $x > 0$ and $y < z$ then $xy < xz$.
- (c) If $x < 0$ and $y < z$ then $xy > xz$.
- (d) If $x \neq 0$ then $x^2 > 0$. In particular, $1 > 0$.
- (e) If $0 < x < y$ then $0 < 1/y < 1/x$.

4 The Real Field

Thm 15. There exists an ordered field $\mathbb{R}$ which has the least-upper-bound property. Moreover, $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield.

Thm 16.

- (a) If $x \in \mathbb{R}, y \in \mathbb{R}$, and $x > 0$, then there is a positive integer n such that

$$nx > y.$$

- (b) If $x \in \mathbb{R}, y \in \mathbb{R}$, and $x < y$, then there exists a $p \in \mathbb{Q}$ such that $x < p < y$.

Thm 17. For every real $x > 0$ and every integer $n > 0$, there is one and only one positive real y such that $y^n = x$.

This number y is written $\sqrt[n]{x}$ or $x^{1/n}$.

Cor 18. If a and b are positive real numbers and n is a positive integer, then

$$(ab)^{1/n} = a^{1/n}b^{1/n}.$$

5 The Extended Real Number System

Def 19. The extended real number system consists of real field $\mathbb{R}$ and two symbols, $+\infty$ and $-\infty$. We preserve the original order in $\mathbb{R}$ and define

$$-\infty < x < +\infty$$

for every $x \in \mathbb{R}$.

6 The Complex Field

Def 20. A complex number is an ordered pair (a, b) of real numbers. "Ordered" means that (a, b) and (b, a) are regarded as distinct if $a \neq b$.

Let $x = (a, b)$, $y = (c, d)$ be two complex numbers, we write $x = y$ if and only if $a = c$ and $b = d$. We define:

$$\begin{aligned}x + y &= (a + c, b + d) \\xy &= (ac - bd, ad + bc)\end{aligned}$$

Thm 21. These definitions of addition and multiplication turn the set of all complex numbers into a field, which $(0, 0)$ and $(1, 0)$ in the role of 0 and 1.

Thm 22. For any real numbers a and b we have

$$(a, 0) + (b, 0) = (a + b, 0), \quad (a, 0)(b, 0) = (ab, 0)$$

Def 23. $i = (0, 1)$.

Thm 24. $i^2 = -1$.

Thm 25. If a and b are real, then $(a, b) = a + bi$.

Def 26. If a, b are real and $z = a + bi$, then the complex number $\bar{z} = a - bi$ is called the *conjugate* of z . The numbers a and b are the *real part* and the *imaginary part* of z , respectively.

We shall occasionally write

$$a = \operatorname{Re}(z), \quad b = \operatorname{Im}(z).$$

Thm 27. If z and w are complex, then

- (a) $\overline{z + w} = \bar{z} + \bar{w}$,
- (b) $\overline{zw} = \bar{z} \cdot \bar{w}$,
- (c) $z + \bar{z} = 2\operatorname{Re}(z)$, $z - \bar{z} = 2i\operatorname{Im}(z)$,
- (d) $z\bar{z}$ is real and positive (except when $z = 0$).

Def 28. If z is a complex number, its *absolute value* $|z|$ is the non-negative square root of $z\bar{z}$; that is $|z| = (z\bar{z})^{1/2}$.

Thm 29. Let z and w be complex numbers. Then

- (a) $|z| > 0$ unless $z = 0$, $|0| = 0$,

- (b) $|\bar{z}| = |z|$,
- (c) $|zw| = |z||w|$,
- (d) $|\operatorname{Re}(z)| \leq |z|$,
- (e) $|z + w| \leq |z| + |w|$.

Notation 30. If $x_1, \dots, x_n$ are complex numbers, we write

$$x_1 + x_2 + \cdots + x_n = \sum_{j=1}^n x_j.$$

Thm 31. If $a_1, \dots, a_n$ and $b_1, \dots, b_n$ are complex numbers, then

$$\left| \sum_{j=1}^n a_j \bar{b}_j \right|^2 \leq \sum_{j=1}^n |a_j|^2 \sum_{j=1}^n |b_j|^2.$$

7 Euclidean Spaces

Def 32. For each positive integer k , let $\mathbb{R}^k$ be the set of all ordered k -tuples

$$\mathbf{x} = (x_1, x_2, \dots, x_k),$$

where $x_1, \dots, x_k$ are real numbers, called the *coordinates* of $\mathbf{x}$. The elements of $\mathbb{R}^k$ are called points, or vectors, especially when $k > 1$. We shall denote vectors by boldface letters. If $\mathbf{y} = (y_1, \dots, y_k)$ and if α is a real number, put

$$\begin{aligned} \mathbf{x} + \mathbf{y} &= (x_1 + y_1, \dots, x_k + y_k) \\ \alpha \mathbf{x} &= (\alpha x_1, \dots, \alpha x_n) \end{aligned}$$

so that $\mathbf{x} + \mathbf{y} \in \mathbb{R}^k$ and $\alpha \mathbf{x} \in \mathbb{R}^k$. This defines addition of vectors, as well as multiplication of a vector by a real number (a scalar). These two operations satisfy the commutative, associative, and distribution laws and make $\mathbb{R}^k$ into a *vector space over the real field*. The zero element of $\mathbb{R}^k$ (sometimes called the *origin* or the *null vector*) is the point $\mathbf{0}$, all of whose coordinates are 0.

We also define the so-called "inner-product" (or scalar product) of $\mathbf{x}$ and $\mathbf{y}$ by

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^k x_i y_i$$

and the *norm* of $\mathbf{x}$ by

$$|\mathbf{x}| = (\mathbf{x} \cdot \mathbf{x})^{1/2} = \left(\sum_{i=1}^k x_i^2 \right)^{1/2}.$$

The structure now defined (the vector space $\mathbb{R}^k$ with the above inner product and norm) is called euclidean k -space.

Thm 33. Suppose $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^k$, and α is real. Then

- (a) $|\mathbf{x}| \geq 0$;
- (b) $|\mathbf{x}| = 0$ if and only if $\mathbf{x} = 0$;
- (c) $|\alpha\mathbf{x}| = |\alpha||\mathbf{x}|$;
- (d) $|\mathbf{x} \cdot \mathbf{y}| \leq |\mathbf{x}||\mathbf{y}|$;
- (e) $|\mathbf{x} + \mathbf{y}| \leq |\mathbf{x}| + |\mathbf{y}|$;
- (f) $|\mathbf{x} - \mathbf{z}| \leq |\mathbf{x} - \mathbf{y}| + |\mathbf{y} - \mathbf{z}|$.

Rmk 34. (a), (b), and (f) will allow us to regard $\mathbb{R}^k$ as a metric space.